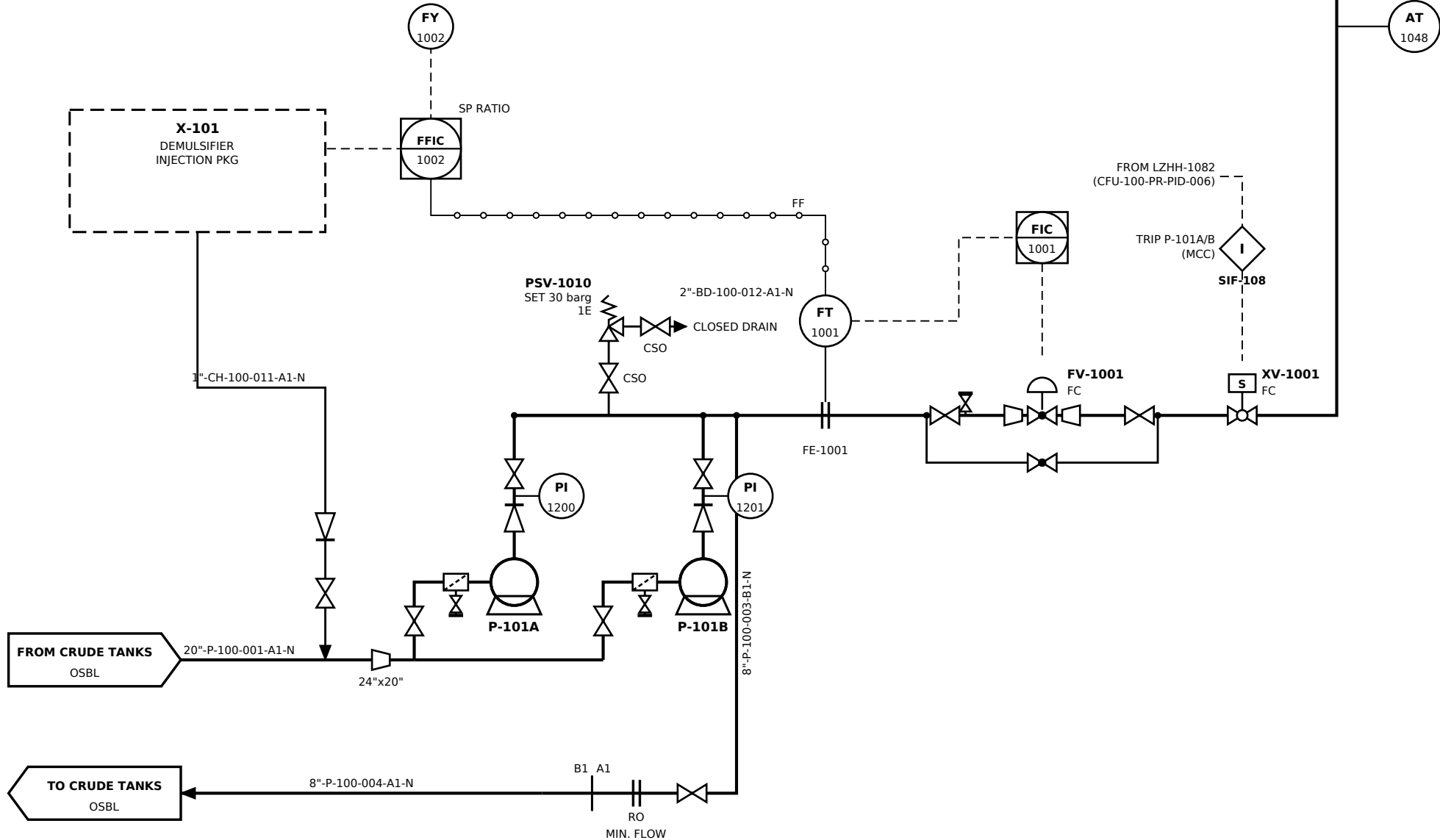
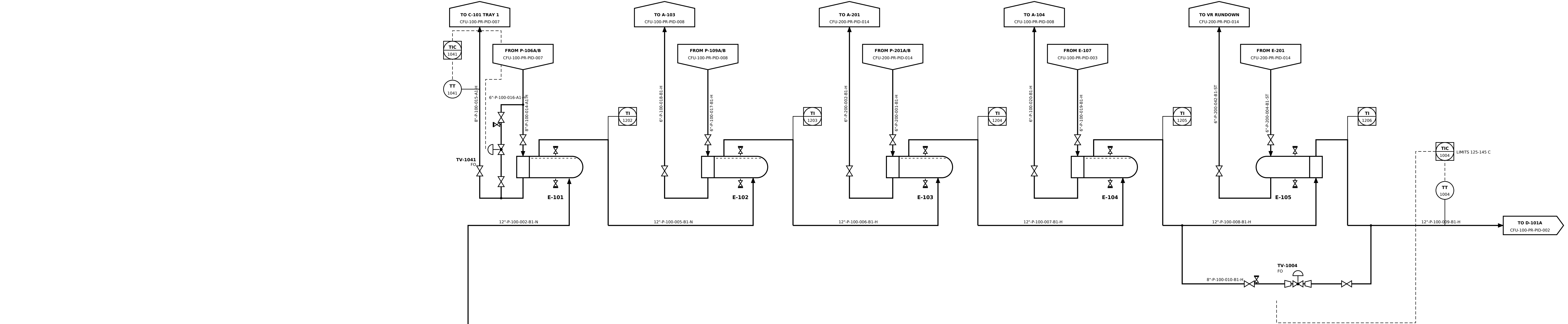


P-101A/B Crude charge RATED 730 m3/h @ 261 m, @P 22 bar MOTOR 710 kW, API 682, Flan 11/22 DES. T 60 C; MAT'L S-6 (CS/12Cr)	E-101 Crude / Top pumparound 359 m2 total; 15 x 1P shells DESIGN 35 / 20 barg / 185 C; CA 3 mm DUTY 8.09 MW SHELL: Crude / TUBE: Top pumparound MAT'L CS shell / CS tubes	E-102 Crude / Kerosene product 438 m2 total; 15 x 1P shells DESIGN 35 / 20 barg / 240 C; CA 3 mm DUTY 6.55 MW SHELL: Crude / TUBE: Kerosene product MAT'L CS shell / CS tubes	E-103 Crude / LVGO pumparound/product 804 m2 total; 25 x 1P shells DESIGN 35 / 20 barg / 255 C; CA 3 mm DUTY 11.45 MW SHELL: Crude / TUBE: LVGO pumparound/product MAT'L CS shell / CS tubes	E-104 Crude / Diesel product 378 m2 total; 15 x 1P shells DESIGN 35 / 20 barg / 235 C; CA 3 mm DUTY 4.44 MW SHELL: Crude / TUBE: Diesel product MAT'L CS shell / CS tubes	E-105 Crude / Vacuum residue 431 m2 total; 15 x 1P shells DESIGN 35 / 20 barg / 285 C; CA 3 mm DUTY 5.84 MW SHELL: Vacuum residue / TUBE: Crude MAT'L CS shell / 5Cr-1/2Mo tubes	X-101 Demulsifier injection package Tank 2 m3 + 2 x 100 % metering pumps MOTOR 0.75 kW MAT'L SS316
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- NOTES**
1. Cold-train crude-side design 35 barg (E-101, E-105) >= P-101 shut-off; PSV-1010 (30 barg) thermal / blocked-in.
 2. Exchanger hot-side connections shown at channel / shell nozzles; TEMA type & shells per data sheets.
 3. TV-1041 / TV-1004 bypass control valves have block valves only (exchanger provides bypass path).
 4. Demulsifier dosed at P-101 suction, ratio to FIC-1001 (FFIC-1002) - X-101 vendor package.
 5. Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
 6. All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
 7. Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST			
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PIPING & INSTRUMENT DIAGRAM			
CRUDE CHARGE PUMPS & COLD PREHEAT TRAIN			
DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	2 OF 17
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-100-PR-PID-001			A